Cooling capacity and SHR

24 unit-specific curve pairs fitted to 1,434 valid catalog points, using the largest listed airflow for each size. 18 manufacturer blanks remain unavailable. The fitted inputs are outdoor dry bulb, entering wet bulb and entering dry bulb; airflow is fixed, not an input.

Three temperature inputs, one fixed airflow selection

Separate fits retain the catalog size and efficiency labels. Selected airflow is 480 CFM per nominal ton in every included table. Source SHR is sensible cooling capacity divided by total gross cooling capacity. Capacity is gross, before indoor fan heat, following the source notes.

Equations - all temperatures in degrees Fahrenheit

```
x = (outdoor DB - 95) / 20

y = (entering WB - 67) / 6

z = (entering DB - 80) / 10

F = 1 + c1*x + c2*y + c3*z + c4*x*x + c5*x*y

+ c6*x*z + c7*y*y + c8*y*z + c9*z*z

Gross capacity (Btu/hr) = Q_reference * F

SHR = min(1, max(0, (b0 + b1*x + b2*y + b3*z) / F))
```

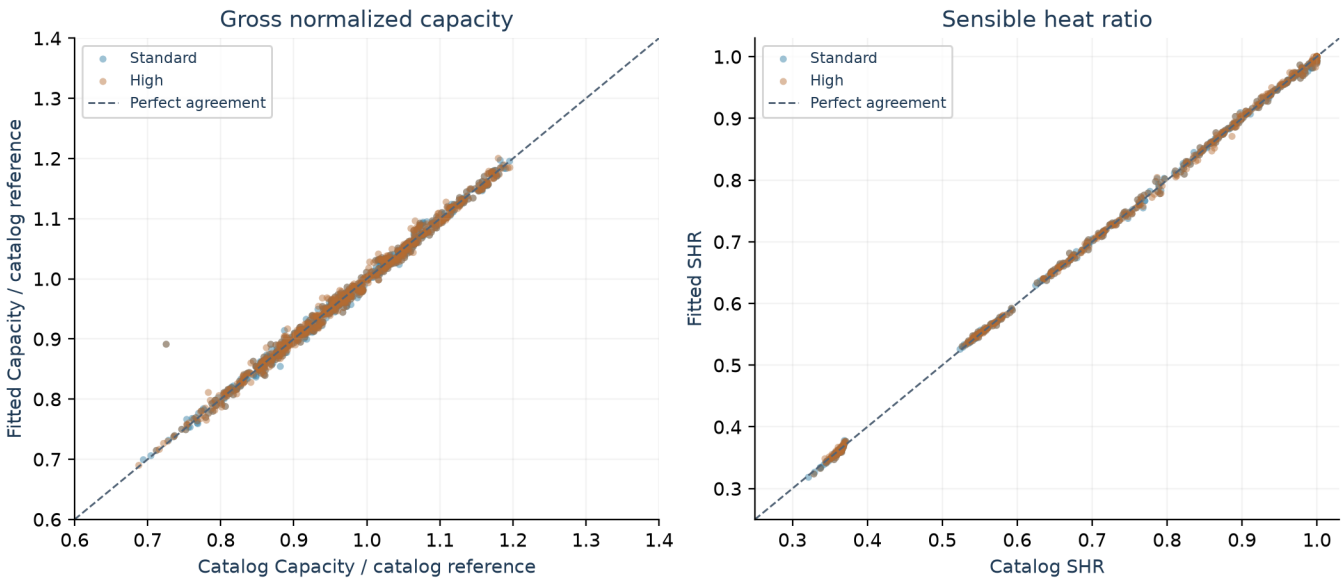
Q_reference is each unit's catalog capacity at 95 F outdoor DB, 67 F entering WB and 80 F entering DB, at the selected maximum airflow. It is neither nominal tonnage times 12,000 nor the peak building load. Capacity is normalized to exactly 1 at that reference. The four b coefficients are part of the SHR function; they introduce no additional operating input.

Where the functions can be used

Entering WB: 61-73 F. Entering DB: 75-90 F. Outdoor DB: generally 85-125 F; standard 10-ton heat pump: 85-120 F. For both 25-ton heat-pump fits, also require outdoor DB + 2 x entering DB <= 295. These limits describe the convex hull of valid catalog points. No extrapolation is allowed.

Coefficients, catalog references, exact domain checks, source rows and validation are provided in the accompanying CSV/JSON files and standalone evaluator. Capacity and SHR fitting is separate from the cooling power and fan balance in the engineering guide.

Accuracy and source review



Each marker compares one catalog point with its fitted result. Dashed lines show perfect agreement. Standard and high-efficiency points often overlap because several source datasets are identical.

Measure	Range across fitted units
Capacity RMSE / reference capacity	0.62% to 2.39%
SHR RMSE (absolute ratio)	0.0014 to 0.0049
Interior-plane validation: capacity RMSE / reference	0.66% to 3.43%
Interior-plane validation: SHR RMSE	0.0016 to 0.0067

Validation method

Capacity uses anchored ordinary least squares. SHR uses bounded-output least squares for its numerator, with the fitted capacity as denominator. Interior-plane validation removes each interior outdoor-DB or entering-DB temperature plane, refits the model, and predicts the omitted points. The reference capacity stays known. Only three wet-bulb levels are supplied, so independent wet-bulb-plane validation is not claimed. These errors measure catalog agreement, not field performance.

Source values requiring review

Both 17.5-ton heat-pump tables show 156.6 MBh at 8,400 CFM, outdoor DB 115 F, entering WB 61 F and entering DB 85 F (printed catalog pages 50 and 61). Visual review confirms that this is printed in the supplied PDF. The value is retained without correction, creates the largest capacity residual, and requires source confirmation. Review flagged-source-values.csv before relying on these two capacity fits.

Source: supplied cooling performance tables; heat-pump tables 10-33 only; dual-fuel tables 34-57 excluded. The selected maximum-airflow data retain PDF and printed page references, manufacturer blanks and all raw values. Source hash is recorded in curve-fits.json.

Unit-by-unit fit summary

Standard efficiency - heat pump

Tons	Fixed CFM	Qref MBh	Cap. RMSE %	SHR RMSE	Table
3	1,440	38.9	0.75	0.0024	10
4	1,920	52.1	0.67	0.0025	11
5	2,400	61.9	0.65	0.0017	12
6	2,880	79.3	0.63	0.0041	13
7.5	3,600	92.0	0.67	0.0014	14
8.5	4,080	110.7	0.62	0.0025	15
10	4,800	124.1	1.07	0.0047	16
12.5	6,000	156.9	0.77	0.0029	17
15	7,200	189.6	0.74	0.0028	18
17.5	8,400	215.9	2.39*	0.0049	19
20	9,600	253.5	0.79	0.0036	20
25	12,000	277.8	0.64	0.0024	21

High efficiency - heat pump

Tons	Fixed CFM	Qref MBh	Cap. RMSE %	SHR RMSE	Table
3	1,440	37.8	1.27	0.0045	22
4	1,920	51.4	0.84	0.0033	23
5	2,400	62.9	0.82	0.0031	24
6	2,880	79.3	0.63	0.0041	25
7.5	3,600	92.0	0.67	0.0014	26
8.5	4,080	110.7	0.62	0.0025	27
10	4,800	124.1	0.78	0.0040	28
12.5	6,000	156.9	0.77	0.0029	29
15	7,200	189.6	0.74	0.0028	30
17.5	8,400	215.9	2.39*	0.0049	31
20	9,600	253.5	0.79	0.0036	32
25	12,000	277.8	0.64	0.0024	33

* Source anomaly retained; see page 2. Capacity RMSE is percent of the catalog reference, not percent of each individual data point. SHR RMSE is an absolute ratio. Per-point errors, relative-capacity errors and held-out predictions are supplied in the CSV files.